DAY 16 | POST-READ

Pipelines and SQL: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Clause order	FROM/JOIN, WHERE, GROUP BY, HAVING, SELECT, ORDER BY/LIMIT.
WHERE vs HAVING	WHERE filters rows; HAVING filters groups after aggregation.
Joins in SQL	INNER for matches only; LEFT to keep all left rows.
CTEs	WITH name AS (...) for readable multi-step queries.
Window functions	RANK, ROW_NUMBER, SUM OVER (PARTITION BY ... ORDER BY ...).
Spark SQL	createOrReplaceTempView then spark.sql(...); returns a DataFrame.
Production pipeline	Config, extract, transform, validate, load and log; parameterised by run date.
Quality checks	Not empty, unique keys, valid ranges, reconciliation. Fail loudly.
Consolidation	Python, APIs, databases, security, React, testing, ETL, Spark: one toolkit.

Part B: Going beyond today

1. ROW_NUMBER for deduplication

```
SELECT * FROM (
  SELECT *, ROW_NUMBER() OVER (PARTITION BY order_id ORDER BY ingested_at DESC) AS rn
  FROM raw_orders
) WHERE rn = 1
```

Keeps only the latest version of each order: the standard way to deduplicate in SQL.

2. LAG for day-over-day change

```
SELECT order_date, revenue,
       revenue - LAG(revenue) OVER (ORDER BY order_date) AS change
FROM daily_total
```

3. Scheduling

Real pipelines run from a scheduler: cron for simple jobs, Apache Airflow or Azure Data Factory for many dependent jobs, or Databricks Jobs. They pass the run date in, retry on failure and alert on errors.

4. Dynamic partition overwrite

```
spark.conf.set("spark.sql.sources.partitionOverwriteMode", "dynamic")
gold.write.mode("overwrite").partitionBy("order_date").parquet(path)
```

Only the partitions present in the new data are replaced, so re-running one day does not delete the others.

5. dbt

dbt lets teams write transformations as SQL SELECT statements with tests and documentation, and builds them in dependency order. It is widely used for ELT in warehouses.

Part C: Check yourself

- Why can you not use a SELECT alias in WHERE?
- When would you use a CTE instead of a subquery?
- What is the difference between RANK and ROW_NUMBER?
- Why should a pipeline take the run date as a parameter?
- What is a reconciliation check?

Part D: Practice ideas for tonight

- Write 5 SQL questions about the Day 16 dataset and answer them with Spark SQL.
- Add a LAG-based day-over-day revenue change to the gold layer.
- Draw a one-page diagram of every tool you have used so far and how they connect.

Coming up on Day 17: DevOps and cloud fundamentals: CI/CD, IaaS, PaaS and SaaS, Azure and Azure DevOps. There is also an assessment (MCQ and coding).

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today